

Cypress Retreat — Design Packet

A 90 ´ × 60 ´ barndominium for Montgomery County, Texas 3 bed / 2.5 bath residence + attached 3-bay activity gym · 5,400 sq ft under roof

Generated with the barndsl architecture DSL. The plan compiles 0 errors; every remaining diagnostic is an intentional design decision or an advisory code note (documented in §8). This packet is a schematic-design study — final geometry, structure, foundation, and materials must be confirmed by a licensed Texas engineer/architect and the local Authority Having Jurisdiction (AHJ).

1. The brief

"A 3 bedroom, 2½ bath, 60 × 90 barndominium featuring a large living room and kitchen with a central island. Tons of storage and a safe room. Attached gym with 3 sections — turf on one side, a rubber-matted tumble area on another, and a boat garage on the outer edge. These areas need safety precautions while maintaining views between spaces so parents can be in a conditioned space. Include climate data for Montgomery County, Texas, and, based on that data, determine the best materials."

2. How the design answers it

Requirement	Where it lives in the plan
3 bedrooms	Primary suite (Mbed, 400 sf) + Bed2 (144 sf) + Bed3 (120 sf)
2½ baths	Mbath (ensuite) + Bath2 (hall) full baths, Half powder bath = 2 full + 1 half
Large living room	Living great room, 560 sf, vaulted, open to kitchen & lounge
Kitchen w/ central island	Kitchen, 16 × 16 (256 sf) — square footprint sized for a large central island + seating
Tons of storage	Walk-in Mcloset (160 sf), two 72 sf bedroom walk-ins, Pantry, Mud/drop-zone, 120 sf bulk Store
Safe room	Safe, 120 sf, dead-center, windowless, hardened (see §5)
Gym — turf	Turf bay, 18 × 30 (540 sf), artificial turf
Gym — tumble	Tumble bay, 18 × 30 (540 sf), rubber matting
Gym — boat garage (outer edge)	Boat, 18 × 60 (1,080 sf) on the far east edge, 12 ´ overhead door
Views + parents in conditioned space	Conditioned supervision Lounge + tempered vision panels chaining all 3 bays (see §5)
Safety precautions	Fire-rated garage separation, self-closing rated doors, CO/smoke alarms, tempered glazing, single controlled house↔gym threshold (see §5)
Montgomery County climate data	§6
Climate-driven materials	§7

3. Program & area summary

Metric	Value
Footprint (under roof)	5,400 sq ft (90 ´ × 60 ´)
Conditioned/interior area	4,320 sq ft
Habitable area	1,680 sq ft
Bedrooms / baths	3 / 2.5
Ceiling	11 ´ typical · great room vaulted · gym bays 14 ´

Metric	Value
Exterior wall area	3,300 sq ft
Roof area (approx.)	5,692 sq ft
Foundation concrete (approx.)	77.8 cu yd
Structural frame	6 bents / 18 posts / 450 ft of beam (18 ' bays, 60 ' clear span)

Full room, door, and window schedules: [schedules.md](#).

4. Plan organization



Architectural floor plan



Floor plan with post-and-beam frame overlay

- Open conditioned core (south): great room ⇌ kitchen ⇌ supervision lounge, one continuous space, with a walk-in pantry and mud drop-zone tucked behind the kitchen.
- Private wing (north/west): the primary suite occupies the full-depth west bay (bed + ensuite + walk-in); a hall spine plus a gallery serve two secondary bedrooms whose walk-in closets stack between them to buffer sound.
- Service core (center): the windowless safe room, powder bath, laundry, and bulk storage cluster on interior walls — no exterior glazing to protect.
- Gym wing (east): three bays in a line — turf and tumble against the house so they can be supervised, the boat garage on the outer (east) edge with its own overhead door.

Drawings: [floorplan.svg](#) (architectural) · [floorplan.png](#) · [floorplan_structural.svg](#) (with the post-frame grid).

5. Supervision & safety systems

Sightlines (the "maintain views" requirement). Parents stay in the conditioned supervision Lounge — part of the open living core, air-conditioned, with seating — and watch the kids play:

- Lounge → Turf: a large tempered vision panel + a single glazed door.
- Turf → Tumble and Turf/Tumble → Boat bay: tempered vision panels create a continuous supervision sightline, so a parent in the conditioned lounge can see across all three activity bays without leaving the a/c.
- All four interior panels are tempered safety glazing (they intentionally trip the DSL's WINDOW_INTERIOR note — that is the brief being satisfied, not a defect).

Controlled threshold. There is exactly one house↔gym door (Lounge→Turf), a self-closing unit, so the conditioned envelope isn't compromised and children pass through a single supervised point. Each gym bay also has its own exterior door for direct outdoor access/egress.

Fire & life-safety (code-driven, flagged by the compiler): - Boat garage ↔ gym common walls detailed as a gypsum fire separation (IRC R302.6; 5/8" Type X where under/adjacent to habitable

space). - Garage-to-room doors are self-closing, 20-minute fire-rated / solid-core (IRC R302.5.1). - Interconnected smoke + CO alarms in every bedroom, outside each sleeping area, and on every level (IRC R314/R315) — required because of the attached garage. - Powder and hall baths are interior → mechanical exhaust vented outside (IRC R303.3).

Activity-area safety detailing (recommended): impact-absorbing wall padding in the tumble bay, poured/rolled shock-attenuating base under the mats, slip-resistant sealed floors, no direct gym-to-bedroom openings, GFCI protection, and separate HVAC/dehumidification zoning for the gym (see §7).

Storm safe room. The central Safe room (120 sf, no exterior walls) should be built to FEMA P-320 / ICC 500 as a residential storm shelter: reinforced cast-in-place concrete or fully-grouted CMU walls, a concrete or steel-plate lid, and an impact-rated door, designed for the hurricane/tornado wind event. It doubles as secure valuables and storm-supply storage.

6. Climate data — Montgomery County, Texas

Montgomery County sits ~40–50 mi north of Galveston Bay (Conroe / The Woodlands / Lake Conroe). It is hot-humid with mild winters, heavy Gulf rainfall, expansive clay soils, and a very high termite load — the material palette in §7 is driven directly by these. (Values are typical design references; verify with a site-specific geotech report and current ASCE 7 / IECC data for the parcel and its FEMA flood zone.)

Parameter	Design value (typical)	Design implication
IECC / ASHRAE climate zone	2A — hot-humid	Cooling- and humidity-dominated; air-tightness + latent control govern
Köppen	Cfa (humid subtropical)	Long cooling season, mild winter
Summer cooling design (1%)	≈ 96–98 °F DB / ~77 °F MCWB	High sensible and latent cooling loads
Winter heating design (99%)	≈ 28–30 °F	Light heating; occasional hard freezes (e.g., Feb 2021)
Degree days	≈ 2,900 CDD65 · ≈ 1,400 HDD65	Cooling >> heating
Mean annual rainfall	≈ 50 in/yr, high-intensity Gulf events	Aggressive roof/wall water management; positive site drainage
Relative humidity	High (frequently 70–90 % AM)	Condensation control on steel; dehumidification
Ultimate design wind (ASCE 7-16, RC II)	≈ 115 mph V _{ult} (≈ 89 mph nominal), Exposure B/C	Engineered uplift/lateral resistance; generally outside the wind-borne-debris region, but hurricane remnants occur (Ike '08, Harvey '17)
Seismic	SDC A (very low)	Not a governing load
Frost depth	Negligible (≈ 0–6 in)	No deep frost footings required
Predominant summer wind	South / southeast (Gulf breeze)	Orient glazing & porches for shade + cross-ventilation
Soils	Expansive clay ("gumbo"), high plasticity/shrink-swell	Engineered foundation is the #1 local risk
Termites	Very high (subterranean + Formosan)	Non-cellulose structure + termite barriers
Flood	Localized — creeks, San Jacinto, Lake Conroe	Confirm FEMA zone; set finished-floor elevation accordingly

7. Climate-driven material & assembly selection

Each recommendation traces back to a climate driver above.

System	Recommendation	Why (climate driver)
Foundation	Engineered stiffened post-tensioned slab-on-grade (PTI DC 10.5) or pier-and-beam with void forms, per geotech; 15-mil under-slab vapor retarder; uniform perimeter moisture (root barriers, drip-free gutters)	Expansive clay shrink-swell + heavy rain — the dominant local failure mode
Primary structure	Galvanized steel post-frame ("red-iron") shell — long clear spans for the gym, pairs with the 18' bays; engineered for ~115 mph V _{ult} uplift/lateral	High wind, big spans, termite-proof, humidity →
Roof	Standing-seam metal, 26-ga Galvalume, light/high-SRI cool-roof color, concealed clips, ~4:12 pitch, continuous ridge vent + uplift straps, ice-&-water at penetrations	Cuts cooling load (hot sun), sheds 50" rain, wind-uplift resistant, no exposed fasteners to leak
Above-deck	Unvented conditioned attic: closed-cell spray foam to the underside of the roof deck; radiant barrier	Keeps ducts in conditioned space; stops condensation on steel in a humid a/c climate
Walls	Metal siding or fiber-cement (James Hardie) over a drainable rainscreen + WRB; closed-cell spray foam in the cavity for air-seal + condensation control	Humidity, wind-driven rain, rot/termite resistance; ccSPF controls the steel dew point
Insulation (Zone 2A targets)	Roof/attic R-38 (or R-30 ccSPF at deck); walls R-13 + continuous; slab perimeter insulation not required in 2A; prioritize air-tightness (blower-door test)	Cooling- and latent-load-dominated climate rewards air-sealing over R-value
Windows / glazing	Double-pane Low-E, SHGC ≤ 0.25, U ≤ 0.30-0.40 (ENERGY STAR South-Central); minimize west glass; deep overhangs/porch shading; tempered at gym panels & near floors/doors	Blocks solar heat gain; shading beats glazing area in a hot climate; safety glazing
HVAC	Right-sized variable-speed heat pump (SEER2 ≥ 15.2) + dedicated/whole-house dehumidification; ducts in conditioned attic; separate zone/mini-split for the gym (turf/tumble semi-conditioned), boat garage vented-only	Latent load control is the hot-humid priority; independent gym zone keeps the residence comfortable
Termite protection	Steel framing is inherently termite-resistant; add slab-penetration termite barriers, borate-treat any wood, bait system, 6" siding-to-grade clearance	Very high SE-Texas termite pressure

System	Recommendation	Why (climate driver)
Interior/wet areas	Mold-resistant gypsum in baths/laundry, sealed/epoxy or polished concrete in gym & boat bay (rubber mats over), floors corrosion-resistant (stainless/hot-dip) fasteners	Humidity, spills, wet activity
Storm hardening	FEMA P-320 safe room (\$5); impact-rated exterior doors; shutter provisions on windward risk openings	Hurricane remnants / tornado

8. Compliance snapshot (barndsl diagnostics)

COMPILE OK — 0 error(s), 4 warning(s), 7 info(s)

Program: 3 bed / 3 bath · 4320 sq ft interior · 1680 sq ft habitable · footprint 5400 sq ft

All rooms are reachable, every bedroom has code egress + a closet, openings fit their walls, and the program matches. The remaining items are intentional and documented:

Code	Count	Status
WINDOW_INTERIOR	4	By design — the tempered parent-supervision vision panels (\$5)
GARAGE_SEPARATION, GARAGE_DOOR	3	By design — fire separation + rated doors already specified (\$5, \$7)
ALARM_CO	1	Reminder — interconnected smoke/CO alarms on the electrical plan (\$5)
BATH_VENT	2	Reminder — exhaust fans for the two interior baths (\$5)
BAY_WIDE	1	Accepted — 18' post-frame bays are normal for a metal barndominium; engineer sizes members

9. Deliverables in this packet

File	Contents
montgomery_barndo.barn	The editable DSL source of record
floorplan.svg / floorplan.png	Architectural floor plan
floorplan_structural.svg	Floor plan with the post-and-beam frame overlay
schedules.md	Room, door, and window schedules
montgomery_barndo.dxf	CAD interchange (23 rooms, 20 openings)
montgomery_barndo.revit.json	Revit exchange for the pyRevit add-in
DESIGN_PACKET.md	This document

10. Suggested next steps

1. Site & FEMA flood-zone check; set finished-floor elevation.
2. Geotechnical report → final foundation design for the expansive clay.
3. Structural engineering of the steel frame for local wind (ASCE 7).
4. MEP design: HVAC load calc (Manual J/S/D) with latent sizing + gym zoning.
5. Confirm the safe room to FEMA P-320 / ICC 500 with the AHJ.
6. Add porches/overhangs on the south & west elevations for shading (recommended).

Schematic study produced with barndsl. Checks are approximate and loosely IRC-based — not a substitute for a licensed designer, engineer, or the AHJ.